Titanic Survival Prediction Using Machine Learning

Internship Project Report

Submitted By

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Domain

Machine Learning Internship

Project Title

Titanic Survival Prediction Using Logistic Regression

1. Objective

The objective of this project is to predict whether a passenger survived the Titanic disaster using machine learning techniques. The model uses passenger information such as age, gender, fare, and passenger class to predict survival.

2. Tools and Technologies Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Joblib

3. Dataset Description

The Titanic dataset contains information about passengers aboard the RMS Titanic.

Dataset Information

- Total Records: 891

- Features: Passenger information
- Target Variable: Survived

Important Features

- Pclass
- Sex
- Age
- SibSp
- Parch
- Fare
- Embarked

Target Classes

- 0 = Did Not Survive
- 1 = Survived

4. Exploratory Data Analysis (EDA)

The dataset was analyzed to understand its structure and identify missing values.

EDA Operations Performed

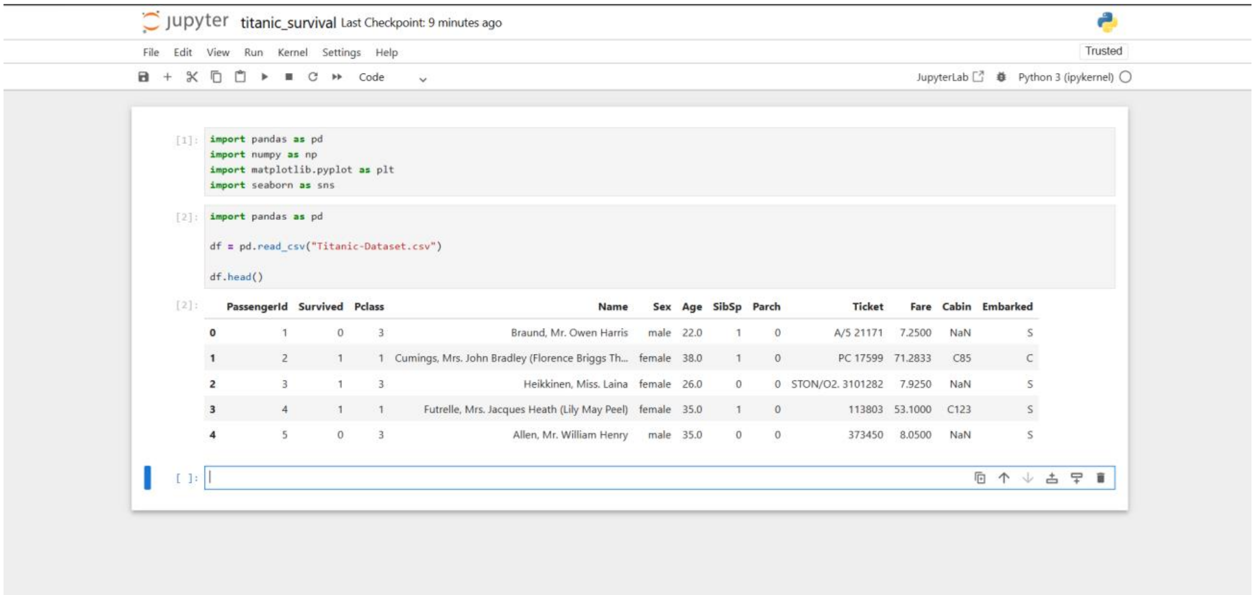
- Displayed first five rows using head()
- Checked dataset information using info()
- Generated statistical summary using describe()
- Identified missing values using isnull().sum()

Observations

- Missing values were found in Age, Cabin, and Embarked columns.
- Missing values were handled using median, mode, and placeholder values.
- Dataset contains both numerical and categorical features.

Screenshots

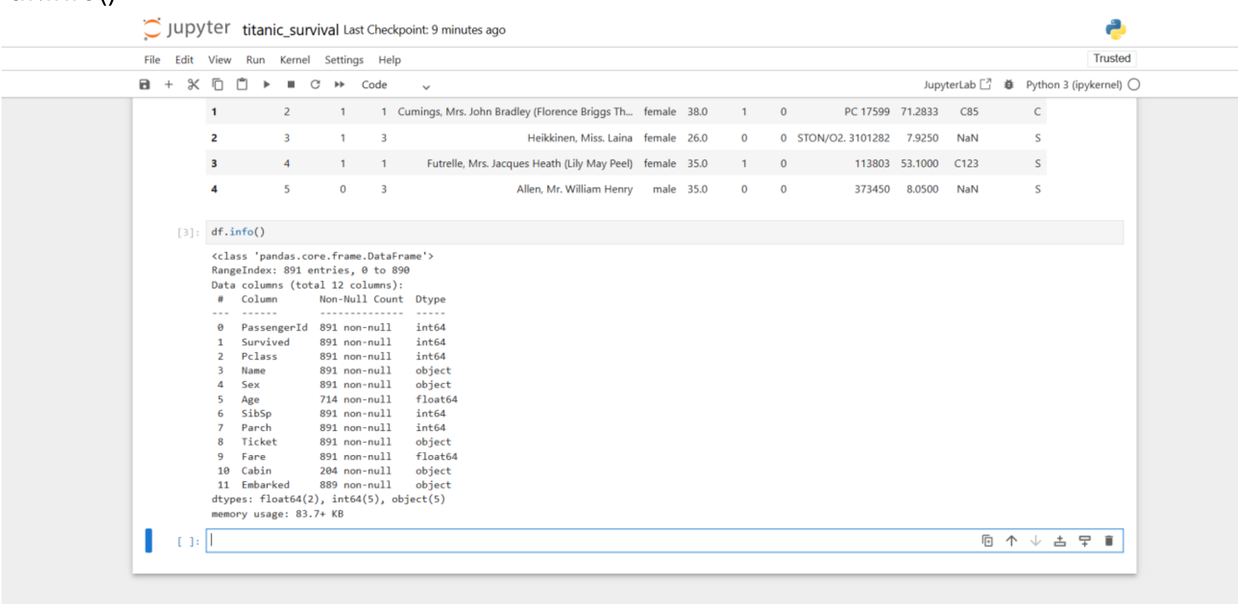
df.head()




```
[1]: import pandas as pd
     import numpy as np
     import matplotlib.pyplot as plt
     import seaborn as sns

[2]: import pandas as pd

     df = pd.read_csv("Titanic-Dataset.csv")

     df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

df.info()



1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
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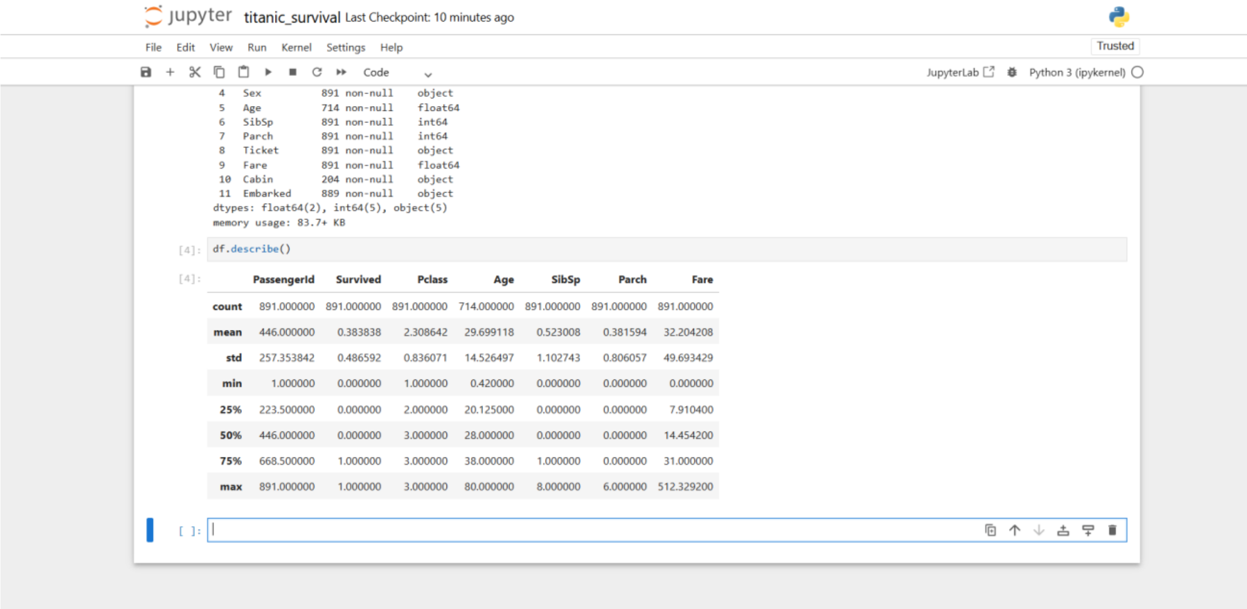
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
[3]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

df.describe()



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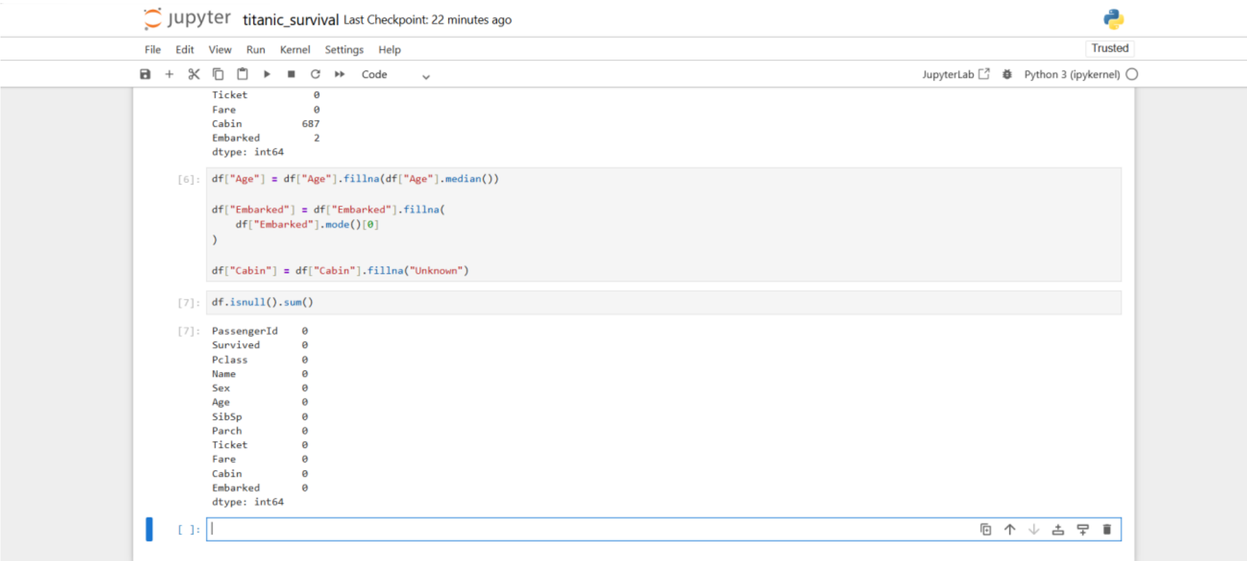
```
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
[4]: df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Missing Values



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```
Ticket        0
Fare          0
Cabin       687
Embarked      2
dtype: int64
```

```
[6]: df["Age"] = df["Age"].fillna(df["Age"].median())

df["Embarked"] = df["Embarked"].fillna(
    df["Embarked"].mode()[0]
)

df["Cabin"] = df["Cabin"].fillna("Unknown")
```

```
[7]: df.isnull().sum()
```

```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin          0
Embarked       0
dtype: int64
```

5. Data Visualization

Several visualizations were created to understand passenger characteristics and survival patterns.

Visualizations Performed

1. Age Distribution Histogram
2. Survival Count Bar Chart

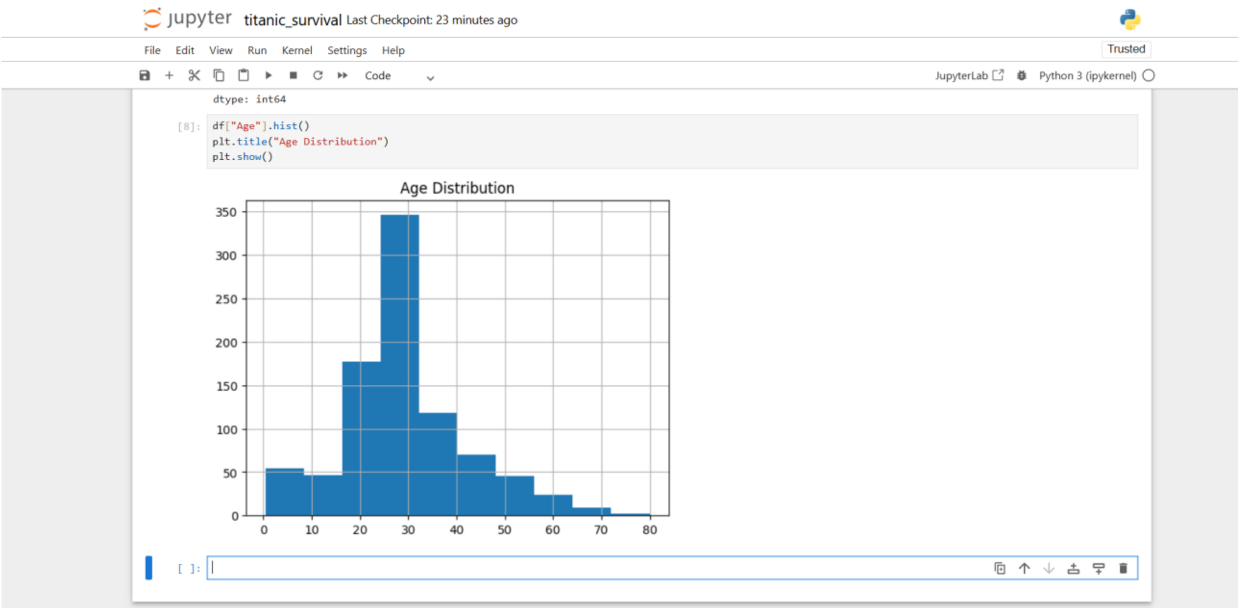
3. Fare Box Plot
4. Age vs Fare Scatter Plot

Observations

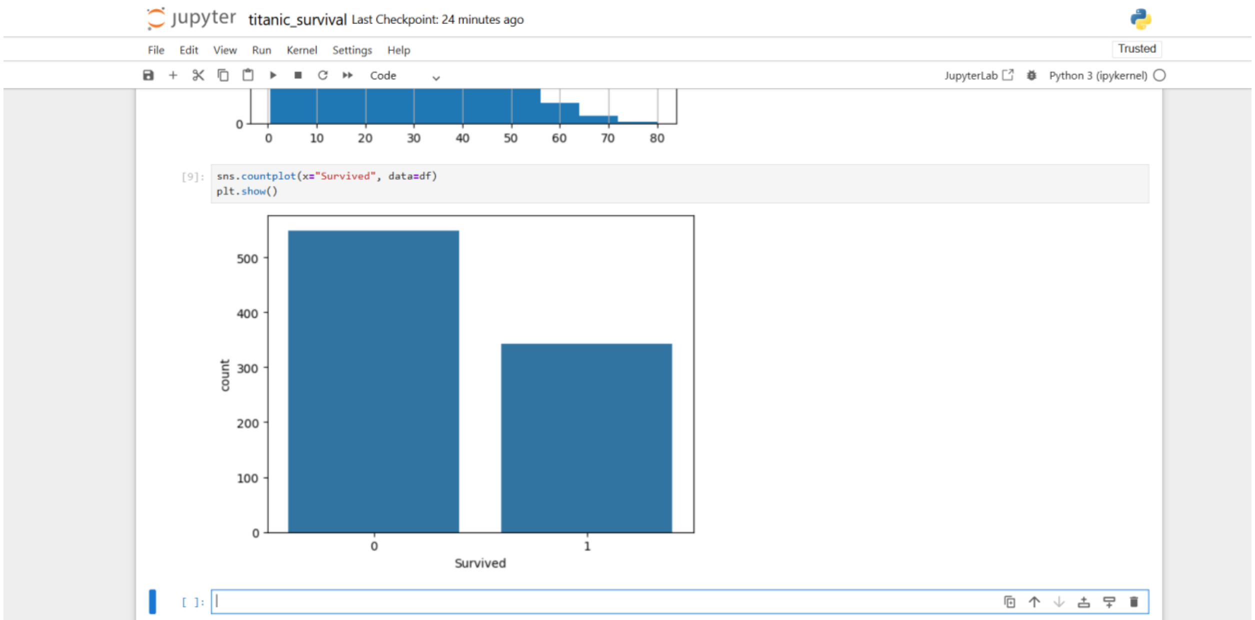
- Most passengers were between 20 and 40 years old.
- More passengers did not survive than survived.
- Fare distribution contained several outliers.
- Passenger age and fare showed varying patterns.

Screenshots

Histogram

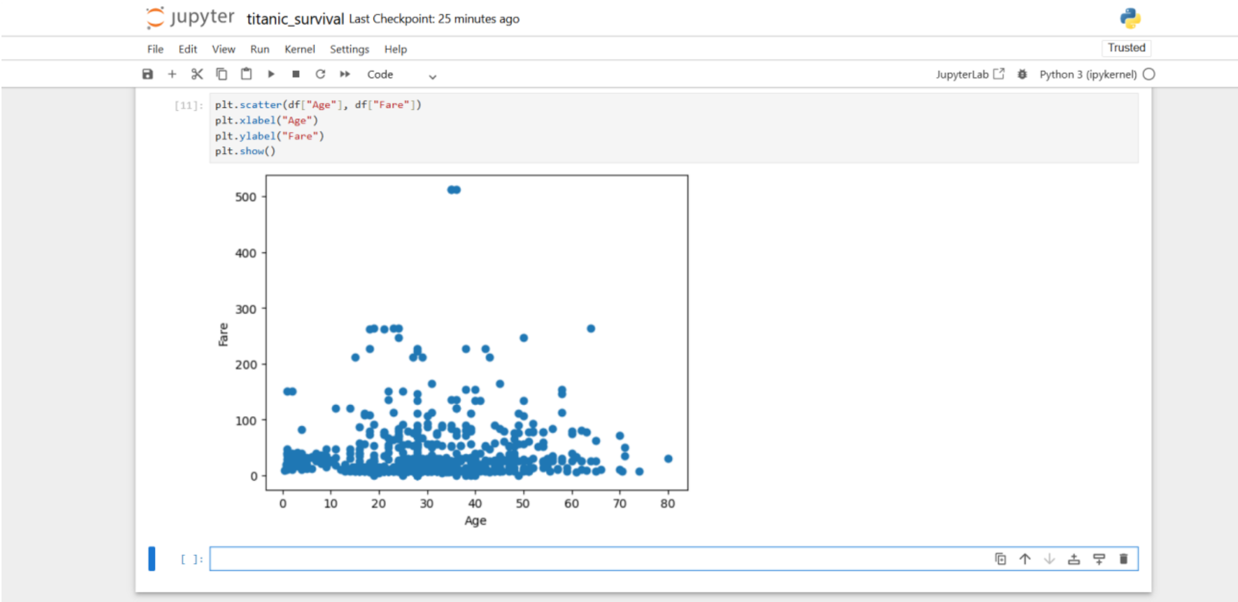


Bar Chart



Box Plot

Scatter Plot



6. Data Preprocessing

Missing Value Handling

- Age column filled using median value.
- Embarked column filled using mode value.
- Cabin column filled with "Unknown".

Feature Encoding

Categorical variables were converted into numerical form for model training.

Examples:

- Male = 0
- Female = 1

7. Model Building

Algorithm Used

Logistic Regression

Steps Performed

1. Data preprocessing

2. Feature selection
3. Train-test split
4. Model training
5. Prediction
6. Performance evaluation

8. Model Evaluation

Evaluation Metrics

- Accuracy Score
- Precision Score
- Recall Score
- Confusion Matrix
- Classification Report

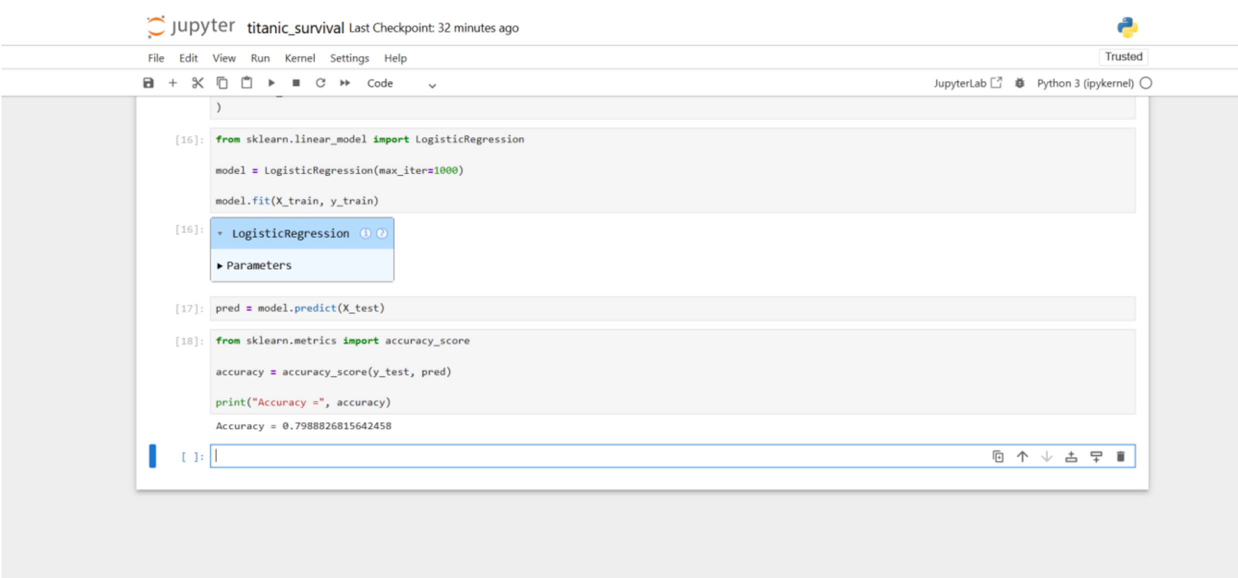
Results

Accuracy: `0.7988826815642458`

Precision: `0.7714285714285715`

Recall: `0.7297297297297297`

Screenshots



9. Model Saving

The trained model was saved using Joblib.

File Name:

titanic_model.pkl

The saved model can be loaded later for prediction without retraining.

10. Conclusion

The Titanic Survival Prediction project was successfully completed using Logistic Regression. The dataset was cleaned, visualized, and used to train a machine learning model. The model was able to predict passenger survival based on available features and achieved satisfactory performance.

This project provided practical experience in data preprocessing, feature engineering, model training, evaluation, and deployment preparation.

11. Repository Artifacts

Files Included:

- titanic_survival.ipynb
- titanic_model.pkl
- titanic_report.pdf
- Screenshots

Submitted as part of the InternSpark Machine Learning Internship Program.